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A STUDY ON UPI ADOPTION IN RAIPUR, CHHATTISGARH

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ABSTRACT:

This study examines the adoption and usage patterns of the Unified Payments Interface (UPI) among users in Raipur, Chhattisgarh. The research aims to identify the key factors influencing adoption, evaluate user satisfaction, and analyze the challenges associated with UPI usage. A quantitative research approach was employed, using primary data collected from 90 respondents through a structured questionnaire.

The findings reveal that UPI adoption is significantly higher among young users, particularly those aged 18–25, indicating the role of digital familiarity in driving usage. Perceived ease of use emerges as the most influential factor, followed by convenience and transaction speed. UPI is primarily utilized for low- to medium-value transactions such as bill payments and online shopping, with a majority of respondents reporting frequent daily usage.

Despite the high adoption rate, users experience challenges such as server downtime and transaction failures, which affect system reliability. While overall satisfaction levels remain high, concerns related to security and technical efficiency persist. The study concludes that although UPI has successfully accelerated the shift toward digital payments, improving infrastructure reliability and enhancing security measures are essential for sustaining long-term user trust and expanding adoption across diverse demographic groups.

Keywords: Unified Payments Interface (UPI); Digital Payments; UPI Adoption; User Satisfaction; Ease of Use; Transaction Security

Introduction

India's financial technology (FinTech) sector has witnessed rapid expansion in recent years, playing a pivotal role in the country's ongoing digital transformation. Within this broader landscape, Tier-II cities such as Raipur have emerged as important contexts for examining how digital financial systems are adopted at the local level. As an administrative and commercial center of Chhattisgarh, Raipur provides a relevant setting to analyze the penetration and usage of digital payment technologies among diverse demographic groups.

One of the most significant innovations in India's digital payment ecosystem is the Unified Payments Interface (UPI), introduced in 2016 by the National Payments Corporation of India (NPCI). UPI enables users to link multiple bank accounts to a single mobile platform, facilitating instant fund transfers, merchant payments, and a wide range of everyday financial transactions. Its simplicity, interoperability, and real-time processing capabilities have significantly enhanced the accessibility and convenience of digital payments.

The adoption of UPI in urban and semi-urban regions such as Raipur has been driven by factors including increased smartphone penetration, improved internet connectivity, and growing awareness of digital financial services. Students, working professionals, and small business owners are increasingly incorporating UPI into their daily financial activities. However, adoption is not uniform across all segments of the population. Demographic variables such as age, education, occupation, and digital literacy continue to influence usage patterns and frequency.

Despite its widespread acceptance, several challenges persist at the local level. These include disparities in digital awareness, concerns related to transaction security, and inconsistencies in system performance across platforms. Such factors shape user perceptions and may affect the sustainability of digital payment adoption over time.

Given these dynamics, this study aims to examine the adoption and usage patterns of UPI in Raipur, with a focus on identifying key drivers, evaluating user satisfaction, and understanding the challenges faced by users. By providing localized, primary data-based insights, the study contributes to a more nuanced understanding of digital payment adoption in emerging urban contexts.

Research Gap

Existing literature on the Unified Payments Interface (UPI) largely focuses on its contribution to Raipur's digital transformation, macroeconomic impact, and theoretical frameworks such as the Technology

Acceptance Model (TAM). Prior studies have identified factors like perceived ease of use, usefulness, and policy support as key drivers of adoption. In addition, research has explored behavioral barriers, security concerns, and the role of UPI in enhancing financial inclusion and supporting the formalization of the informal sector.

However, there is limited empirical evidence that reflects region-specific user experiences, particularly at the city level such as Raipur. Most existing studies are based on national-level data, which may not fully capture local variations in usage patterns, user satisfaction, and challenges faced by individuals.

Moreover, while theoretical models explain adoption behavior, there remains a gap in understanding how these factors function in real-life contexts among specific groups, especially students and young users. Therefore, this study seeks to address this gap by providing primary data-based insights from respondents in Raipur, focusing on actual usage behavior, satisfaction levels, and the challenges associated with UPI adoption.

Research Objectives

- For analyzing the present day presence of UPI services and for identifying the patterns of growth of UPI services based on the transaction volume among different demographics in Raipur.
- Identifying the factors behind the adoption of UPI services and exploring the effect of incentives from the government and “JAM” trinity in motivating customers to move away from traditional payment methods to UPI services.
- Determining the reasons hindering the universal adoption of UPI services: Identifying the constraints involved in the adoption of the UPI service like lack of digital literacy, security issues.

Review of Literature

• Behavioral and Psychological Obstacles

While macro-level data indicates significant growth in UPI adoption, a substantial body of literature examines the underlying reasons for hesitation among certain population groups. This phenomenon, often referred to as “post-adoption dissonance,” has been discussed in studies by Kumari and Meganathan (Kumari, 2023; Meganathan, 2023). In contrast to younger, urban users who readily adopt UPI due to its convenience, rural and older populations tend to experience psychological barriers.

Research also highlights the presence of emotional and cultural factors influencing payment preferences.

In certain social and cultural contexts, physical cash continues to hold symbolic importance, which may limit the acceptance of digital payment methods (EELET, 2023).

Furthermore, “security anxiety” is identified as a key behavioral constraint. Concerns related to social engineering frauds, along with frequent technical issues such as server downtime, contribute to a perceived lack of reliability. This combination of factors creates a “trust deficit,” which discourages some users from fully transitioning to digital payment systems (Kumari, 2023).

- **Government Policy and the Global Expansion Framework**

The Government of India’s Digital India initiative has played a significant role in strengthening the digital payments ecosystem. The JAM Trinity (Jan Dhan, Aadhaar, and Mobile) has provided a foundational framework that has facilitated the expansion of UPI services across the country. Recent reports indicate a positive relationship between the increase in Jan Dhan accounts and the growth of UPI adoption, particularly in rural areas (Press Information Bureau, 2026; Choudhary & Mishra, 2023).

Furthermore, the integration of Aadhaar-linked bank accounts with mobile devices has simplified the onboarding process for users, thereby reducing barriers to adoption and enhancing accessibility to digital payment systems (Press Information Bureau, 2026).

- **Application of the Technology Acceptance Model (TAM)**

Based on the survey conducted among respondents in Raipur the Technology Acceptance Model (TAM) posits that "Perceived Ease of Use" and "Perceived Usefulness" are the two principal factors influencing an individual's decision to adopt new technology.

Perceived Ease of Use (PEOU)

Perceived Ease of Use denotes the extent to which an individual believes that utilizing a specific system would require minimal effort.

- **Data Support:** A notable 44.4% (40 respondents) identified "Ease of Use" as the principal factor affecting their satisfaction.
- **User Motivation:** Furthermore, 77.8% (70 respondents) cited "Ease of Use" as their primary incentive for utilizing UPI applications.
- **Application:** The research indicates a significantly elevated perceived ease of use (PEOU) for the Unified Payments Interface (UPI). The interface's simplicity enables users to execute transactions with minimal cognitive effort, which is the primary factor influencing adoption in the sample.

Perceived Usefulness (PU)

It refers to the extent to which an individual believes that utilising a specific system will improve their job performance or daily activities.

- **Data Support:** Users predominantly employ UPI for fundamental daily activities: 32.2% (29 respondents) for bill payments and 31.1% (28 respondents) for online shopping.
- **Efficiency:** Additionally, 31.1% (28 respondents) indicated that "Speed of Transaction" is their primary element for satisfaction.
- **Application:** The results suggest elevated PU levels. UPI is regarded not merely as a "novel instrument" but as a valuable utility that enhances the efficiency and convenience of financial administration relative to conventional cash or banking procedures.

External Variables and Perceived Risk

TAM frequently considers extrinsic obstacles that may impede the shift from "Intention to Use" to "Actual Usage."

- **Identified Barriers:** Despite elevated perceptions of ease of use (PEOU) and usefulness (PU), 62.2% (56 respondents) experience "Server Downtime," while 60% (54 respondents) encounter "Transaction Failures."
- **Security Concerns:** Merely 51.1% (46 respondents) report feeling "Secure" while utilising the applications, whilst 48.9% (44 respondents) indicate a necessity for "Enhanced Security Features" for improvement.
- **Applications:** In research, "Perceived Risk" (including security concerns and technical malfunctions) functions as a detrimental moderator. Although consumers perceive the system as user-friendly and beneficial, these technical hazards inhibit their complete reliance on UPI for all transactions.

Research Methodology

• Research Design

The basis of the framework underlying this research is the Descriptive and Analytical Research Design that is most suitable for the investigation of the various influences on the usage of the UPI in Raipur. It was deemed that a descriptive design would be the most appropriate method, as it helps to depict the features of the studied population, in particular, their usage of digital payments, without changing the environment. Besides, there is an analysis part aimed at understanding not just the numbers but why users conduct certain actions and how demographic factors affect the number of digital transactions.

The Quantitative Research Paradigm will be applied in this research as well. It involves using numeric data for analyzing user's opinions about certain technologies and determining the level of readiness for their usage. Moreover, it is a cross-sectional type of design that allows collecting information at one moment in order to give a status report on the FinTech field in Raipur in 2025-2026

- **Data Sources**

In order to make the findings from the research reliable and valid, there has been a dual-level approach towards the data acquisition. Both primary and secondary streams of data were used to get the relevant information.

- **Primary Data Collection:** Primary data is the first-hand data needed for the research. It has been collected with the help of an original field survey. In this regard, an electronic questionnaire has been used to get access to the "real-life" experience of UPI users and collect data related to personal age-related use and local barriers. It was considered vital since the researcher needed to answer very specific research questions that could not be covered by national statistics.
- **Secondary Data Collection:** Secondary data was used to develop the contextual background of the study and to support the interpretation of primary survey findings related to UPI adoption in Raipur. The data was collected from reliable and credible sources to ensure academic validity and conceptual clarity. Here are the secondary sources:
 - Institutional Reports: The macro-economic and historical trends related to transactions in India were extracted from official publications and annual statistical handbooks of NPCI and RBI.
 - Global Consultancy Reports: The reports generated by PwC India, Deloitte, and NASSCOM were used to gather information regarding the future trajectory of the technology innovations of the FinTech industry in India.
 - Academic Peer Reviewed Journals: Articles published in *the International Journal of Creative Research Thoughts* (IJCRT) and *TIJER* were used for gaining the theoretical insight in terms of socio-technical theories of technology adoption, such as the TAM model.
 - Government Policy Portals: Reports by the PIB and MeitY were utilized to gain information on the policy-based catalysts, including the 'Digital India' initiative.
- **Sampling Technique**

The research employs Convenience sampling, a non-probability sampling technique, to collect primary data from 90 participants.

- **Justification for Selection:** This strategy was chosen primarily because of the significant accessibility of the target demographic within the academic and local community context. In light of the swift transformation of the FinTech sector in 2026, convenience sampling facilitated a time-efficient and economical data collection amidst significant transaction volatility. Moreover, as the research centres on smartphone-integrated technology, sampling from a recognized tech-savvy environment guaranteed an elevated response rate and superior data quality concerning specific app features.

The use of convenience sampling yielded prompt and practical data; however, it is recognized that this approach introduces specific constraints on the study's external validity. The sample is predominantly composed of individuals aged 18–25 (78.9%) and undergraduate students (63.1%), which may limit the generalizability of the findings to the broader population of Raipur, especially among the elderly or those residing in remote rural areas. This "selection bias" is mitigated by seeing the survey as a focused examination of "Digital Natives" rather than a comprehensive national census.

- **Research Instrument**

The principal instrument used for the collection of primary data was a questionnaire that was made electronically through Google Forms. The questionnaire was categorized into three categories, namely:

Demographic profile: The questions in this category related to the Age, Gender, Place of Residence (Urban/Rural) and Occupation of the respondents. These variables are extremely important in establishing the presence of the 'Digital Divide' in literature.

Data Analysis

Table No. 1: Result of Questionaries

Sl. No	Variable	Category	Frequency	Percentage
1	Age Group	Below 18	7	7.8%
		18–25	71	78.9%
		26–35	8	8.9%
		35–50	4	4.4%
2	Gender	Male	28	31.1%
		Female	62	68.9%
		Other	0	0%

		Prefer not to say	0	0%
3	UPI Applications Used	Google Pay	35	38.9%
		PhonePe	67	74.4%
		Paytm	16	17.8%
		BHIM	6	6.7%
		Other	5	5.6%
4	Frequency of Using UPI	Multiple times a day	29	32.2%
		Daily	36	40.0%
		Weekly	13	14.4%
		Occasionally	12	13.4%
5	Factors Influencing Satisfaction	Ease of use	40	44.4%
		Speed of transaction	28	31.1%
		Security	17	18.9%
		Cashback	5	5.6%
6	Issues Faced Using UPI	Transaction failures	54	60.0%
		Delayed refunds	14	15.6%
		Server downtime	56	62.2%
		Frauds	7	7.8%
		OTP issues	8	8.9%

Analyzing and Interpreting Data

- Demographic Profile: Age Group**

Figure No. 1 shows the significant concentration of users among the 18–25 ages demographic indicates that UPI use is most prevalent among Generation Z and young millennials. This corresponds with the Technology Acceptance Model (TAM), since younger populations generally have elevated "Perceived Ease of Use" and are more inclined to accept digital innovations earlier than older age cohorts.

What is your age group?

90 responses

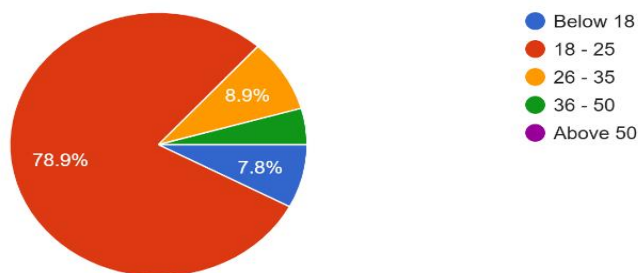


Figure No. 1: Age Group of Respondents

- **Demographic Profile: Gender**

The Figure No. 2 reveals a greater involvement rate among female respondents in this study. This demographic division is crucial for assessing whether gender influences "Perceived Risk," however the elevated overall satisfaction rates indicate that UPI has evolved into a gender-neutral utility in urban and semi-urban environments.

What is your gender?

90 responses

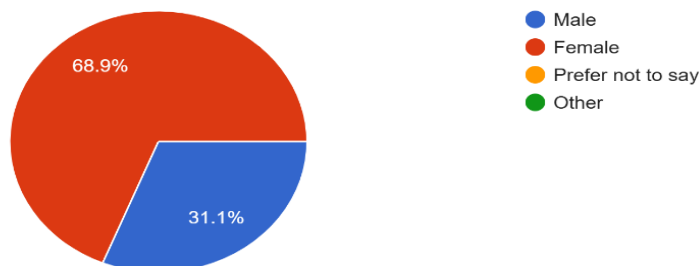


Figure No. 2: Gender of the Respondents

- **Primary UPI Applications Used**

Figure No. 3 shows that PhonePe and Google Pay unequivocally lead in market share within the assessed age. The preference for PhonePe indicates that its user interface and effective merchant integration offer greater "Perceived Usefulness" to consumers than the original government-supported BHIM app.

Which UPI applications do you primarily use?

90 responses

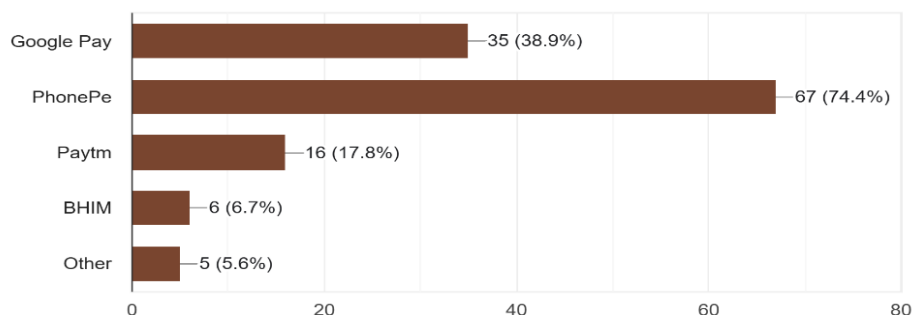


Figure No. 3: Applications used by Respondents

- Factors Influencing Satisfaction**

Figure No. 4 shows that these data fully corroborate the TAM paradigm, wherein "Perceived Ease of Use" serves as the primary factor of user happiness. Although security is a problem, the operational speed and simplicity of the interface are the primary factors that keep the user base.

How would you rate your overall satisfaction with UPI services?

90 responses

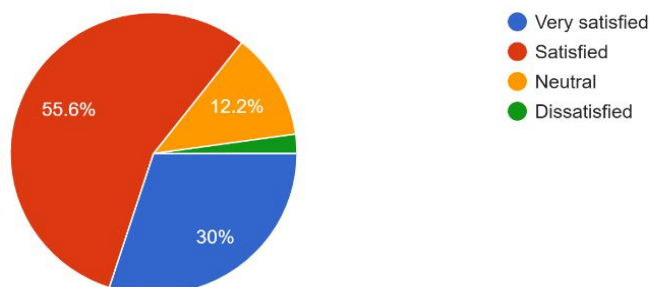


Figure No. 4: Factors influencing satisfaction of Respondents

- Daily Usage Frequency**

Figure No. 5 reveals that with more than 72% of the sample utilizing UPI daily, the technology has evolved from a "novelty" to a "essential." The elevated frequency indicates a robust "Behavioral Intention" to maintain a cashless approach, since the system has effectively assimilated into the respondents' daily routines.

On average, how frequently do you use UPI for transaction?

90 responses

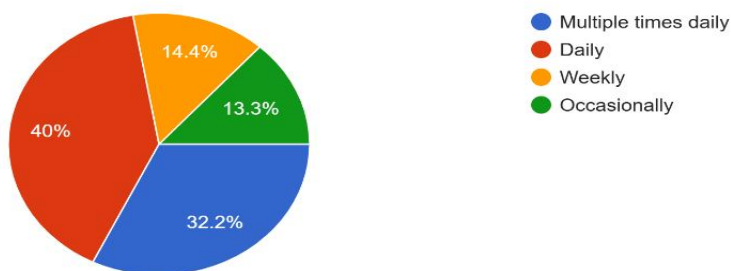


Figure No. 5: Daily Usage Frequency Respondents

• User Challenges

The Figure No. 6 reveals a discrepancy between "Ease of Use" and "System Reliability." Although consumers desire to utilize the system, the infrastructure (server/bank uptime) constitutes the most vulnerable aspect. To achieve general adoption of UPI, the emphasis must transition from user acquisition to the resilience of technical infrastructure.

What kind of issues have you faced while using UPI?

90 responses

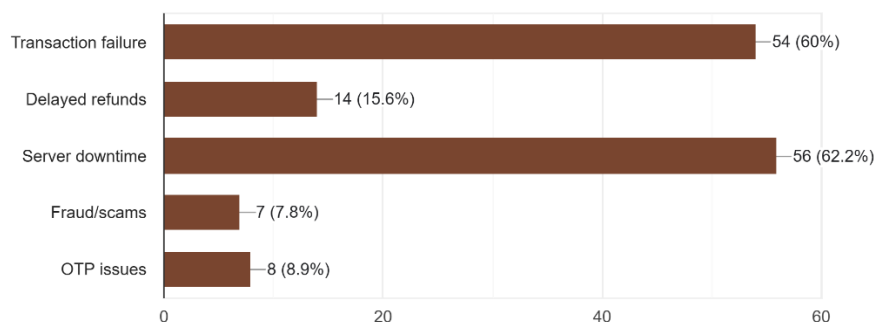


Figure No. 6: Issues faced by Respondents

• Cross-Tabulation Analysis

(A) Age × Usage Frequency

Table No. 2: Cross-Tabulation Results for Age and Usage Frequency

Age Group	Multiple Times/Day	Daily	Weekly	Occasionally	Total
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Below 18	2	3	1	1	7
18–25	24	28	10	9	71
26–35	2	4	1	1	8
35–50	1	1	1	1	4
Total	29	36	13	12	90

Interpretation

The cross-tabulation of Table No. 2 indicates that respondents aged 18–25 constitute the largest group and exhibit the highest frequency of UPI usage, particularly on a daily and multiple-times-per-day basis. In contrast, respondents from other age groups show comparatively lower and more evenly distributed usage patterns. Despite this variation, UPI usage is observed across all age categories, suggesting its broad accessibility. However, the higher concentration of frequent usage among younger respondents reflects greater familiarity and engagement with digital payment systems within this group in Raipur.

(B) Gender × Satisfaction Factor

Table No. 3: Cross-Tabulation Results for Gender and Satisfaction Factor

Gender	Ease of Use	Speed	Security	Cashback	Total
Male	12	9	5	2	28
Female	28	19	12	3	62
Total	40	28	17	5	90

Interpretation

The Table No. 3 shows that ease of use is the most important factor influencing UPI usage for both male and female respondents, accounting for the highest number of responses. This is followed by speed of transaction, which is also a significant consideration across both groups. Security is relatively less cited, while cashback appears to have minimal influence on user preference. The pattern is consistent across genders, indicating that functional aspects such as convenience and efficiency play a more important role than promotional incentives in influencing UPI adoption among respondents in Raipur.

• Chi-Square Test

Test 1: Age Group × Usage Frequency

Hypotheses:

- Null Hypothesis (H_0): There is no significant relationship between Age Group and Frequency of UPI usage.
- Alternative Hypothesis (H_a): There is a significant relationship between Age Group and Frequency of UPI usage.

Table No. 4: Chi-Square Test Results for Age Group and Usage Frequency

Statistical Measure	Value
Pearson Chi-Square (χ^2)	2.454
Degrees of Freedom (df)	9
Asymptotic Significance (p-value)	0.982
Sample Size (N)	90

Interpretation

In Table No. 4 Pearson Chi-Square value of 2.454 with 9 degrees of freedom yields a p-value of 0.982, which is significantly higher than the standard significance level of 0.05. Therefore, the null hypothesis cannot be rejected. This indicates that there is no statistically significant association between the variables under study. The results suggest that any observed differences are likely due to chance rather than a meaningful relationship. Hence, the variables appear to be independent within the sample.

Test 2: Gender $\times$ Satisfaction Factor

Hypotheses:

- Null Hypothesis (H_0): Satisfaction factors are independent of Gender.
- Alternative Hypothesis (H_a): Satisfaction factors are significantly associated with Gender.
-

Table No. 5: Chi-Square Test Results for Gender and Satisfaction Factor

Statistical Measure	Value
Pearson Chi-Square (χ^2)	0.244
Degrees of Freedom (df)	3
Asymptotic Significance (p-value)	0.970
Sample Size (N)	90

Interpretation

In Table No. 5 the Pearson Chi-Square value of 0.244 with 3 degrees of freedom results in a p-value of 0.970, which is considerably higher than the 0.05 significance level. Therefore, the null hypothesis cannot be rejected. This indicates that there is no statistically significant association between the variables under study. The findings suggest that any observed variation is likely due to random chance, and the variables are independent within the sample.

Ethical Considerations

This study ensures that all ethical standards related to research involving human respondents were duly followed. Prior to data collection, informed consent was obtained from all participants, and they were clearly informed about the purpose of the study and the voluntary nature of their participation. Respondents were not subjected to any form of pressure or obligation to participate.

The study also maintains strict anonymity of respondents, ensuring that no personal identifiers such as names, contact details, or sensitive information were collected or disclosed at any stage of the research. This was done to protect the privacy of individuals and encourage honest responses.

Furthermore, data confidentiality was strictly maintained. The information collected through the questionnaire was used solely for academic purposes and was not shared with any third party. All responses were analyzed in aggregated form to ensure that individual responses could not be traced back to any participant.

Findings and Suggestions

The findings of the study indicate that the Unified Payments Interface (UPI) has become an essential mode of digital payment among respondents in Raipur, particularly among the younger population. The adoption of UPI is primarily driven by its convenience, ease of use, and time efficiency rather than short-term incentives such as cashback offers. This suggests that users perceive UPI as a practical and reliable payment method for everyday transactions.

Another important outcome is the variation in usage patterns across different demographic groups. Younger users, especially females in the sample, tend to use UPI more frequently compared to other groups. However, despite the overall positive perception and widespread usage, users continue to face technical challenges such as server downtime and transaction failures.

This reflects a gap between user acceptance and system reliability. While users trust and regularly use UPI services, technical issues still affect their overall experience. Therefore, improving the reliability and performance of the system is essential to sustain and further enhance user adoption.

Suggestions and Recommendations

Based on the findings of the study, several measures can be suggested to improve the effectiveness and adoption of UPI services among users in Raipur. Firstly, there is a need to enhance the reliability of the backend infrastructure managed by the National Payments Corporation of India and participating banks. Frequent issues such as server downtime and transaction failures should be addressed to ensure a smoother user experience.

Secondly, financial institutions should focus on improving grievance redressal mechanisms. A faster and more transparent resolution system can increase user confidence and satisfaction with UPI services.

Further, the introduction and wider adoption of features such as low-value transaction systems (e.g., UPI Lite) can help reduce the load on core banking systems and improve transaction success rates.

For small businesses and local merchants, integrating basic digital record-keeping features within UPI applications can support better financial management and improve access to formal credit facilities.

Lastly, efforts should be made to address demographic barriers by promoting digital literacy and developing simplified user interfaces, especially for elderly and less digitally literate users.

Conclusion

The findings of this study indicate that the Unified Payments Interface (UPI) has become a widely adopted and integral component of digital payment behavior among users in Raipur, particularly within the younger demographic. The results highlight a clear transition from traditional cash-based transactions to digital modes, driven primarily by factors such as ease of use, convenience, and transaction efficiency.

The study also demonstrates that UPI is not merely a supplementary payment option but has evolved into a routine financial tool embedded in users' daily activities. Additionally, it contributes to broader financial

inclusion by enabling small businesses and individual users to participate more actively in the formal financial ecosystem.

However, despite strong adoption and positive user perception, certain challenges remain. Technical issues such as server downtime and transaction failures continue to impact user experience and may influence long-term trust in the system. Furthermore, the study acknowledges limitations in terms of sample representation, as the findings are largely reflective of young, digitally literate users.

In conclusion, while UPI has achieved substantial success in promoting digital payments, its sustained growth will depend on strengthening technological infrastructure, improving security frameworks, and addressing demographic disparities through enhanced digital literacy initiatives. These measures are essential to ensure broader, more inclusive adoption and to reinforce user confidence in the evolving digital payment landscape.

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